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A comprehensive analysis of machine learning-based intrusion detection systems: evaluating datasets and algorithms for internet of things

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ABSTRACT

With the recent advancement of the Internet of Things (IoT) in various sectors, security has become an essential requirement. Any IoT application or device may be compromised by intruders to disrupt the entire network. These kinds of insider attacks are difficult to prevent. Here, an Intrusion Detection System (IDS) can play an important role in identifying various unknown attacks. IDS uses network traffic logs to detect and respond to suspicious activities or anomalies before attackers exploit system weaknesses. Machine learning models are among the most efficient and effective methods to identify anomalous network behaviors. Hence, in this paper, we have conducted a comprehensive analysis utilizing several supervised and semi-supervised machine learning algorithms to assess their performance. We utilized 15 benchmark datasets containing network traffic samples related to various network attacks. We employed holdout and k-fold cross-validation methods for performance comparison. We also discussed the performance of the algorithms and identified possible reasons for their respective outcomes. Experimental results indicate that two supervised algorithms, kNN and ANN, exhibit the highest performance in terms of accuracy, precision, recall, etc. This comprehensive analysis of datasets and algorithms with various evaluation metrics provides researchers with valuable insights.

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IoT; IDS; ML; attacks; security

1. Introduction

Internet of Things (IoT) is a revolutionary technology where objects from different domains are interrelated and connected with each other over the internet. As soon as businesses began using IoT infrastructure, the possibility of security flaws surfaced. Organizations have implemented a variety of information security measures to prevent intruders from gaining access to private

information or damaging the IoT network [1–3]. Intrusion detection systems (IDS) are a crucial part of any comprehensive cybersecurity strategy. With the information it provides, an IDS may spot the suspicious data flow and trigger an alarm for the network admin. The four functional modules that make up an IDS's architecture are the 'E' (Event) block, the 'D' (Database), the 'A' (Analysis) and the 'R' (Response) block. Simply stated, the E block (Event-Boxes) is a collection of sensors used to keep an eye on the system and collect data on occurrences for further analysis. The information gathered must be saved before it can be processed. Information from the E block is saved in the components of the D block (Database boxes). When looking for harmful activity, the events are analyzed in a processing module called A block (Analysis-boxes). The primary objective, after the identification of suspicious actions, is to eliminate the risk. Therefore, the R block (Response boxes) will take the initiative to counteract the incursion with the correct response execution if and when it happens. An illustration of the IDS framework is given in [Figure 1](#)

An Intrusion Detection System (IDS) may be divided into subcategories known as Host-based IDS (HIDS) and Network-based IDS (NIDS), depending on the information source (E block). Network intrusion detection systems (NIDS) analyze network events, such as IP addresses, protocols, service ports and traffic volume. In contrast, host intrusion detection systems (HIDS) focus on information linked to operating systems, such as system calls and process IDs.

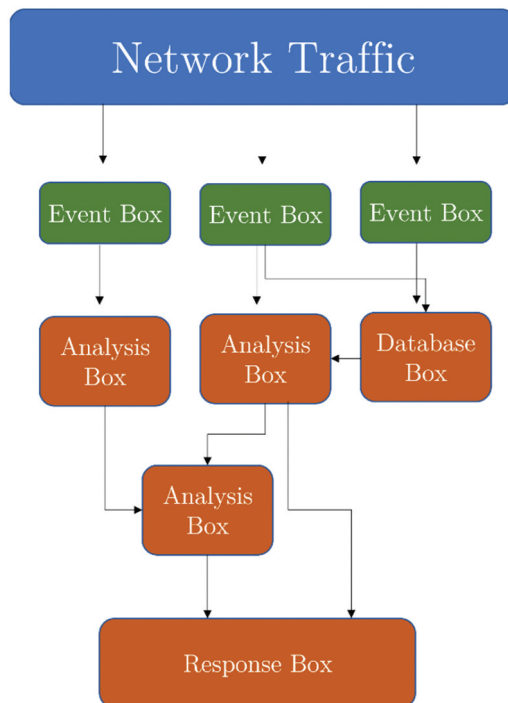


Figure 1. Conventional architecture of IDS.

Signature-based IDS (misuse-based) and anomaly-based IDS categorize IDS based on the analysis conducted in the A block. To detect an intrusion, Signature-based IDS (SIDS) compare examined data to a database of known attack signatures [4]. While proficient at identifying common threats, it is blind to emerging ones. Signature-based intrusion detection systems exhibit a significantly reduced false-positive rate. Anomaly-based intrusion detection systems (AIDS) work to establish a baseline of acceptable system behavior. An anomaly alert is triggered when the observed value at a particular time deviates from the expected value by more than a threshold value [5]. Anomaly-based intrusion detection systems are useful for identifying new types of attacks, as they focus on anomalous behavior. Nonetheless, AIDS has a higher false-positive rate of intrusion detection compared to SIDS.

The exponential growth in cloud service and IoT adoption has led to a corresponding increase in network traffic. IoT devices often lack robust inbuilt security, making them easy targets for cyberattacks such as Distributed Denial of Service (DDoS) [6,7], data breaches [8] or ransomware [9]. IDS monitors network traffic and device behavior in real-time to detect and mitigate suspicious activities or anomalies, preventing attackers from exploiting system weaknesses [10,11]. By employing lightweight [12] and efficient IDS, IoT systems can ensure data integrity, confidentiality and system availability, which are crucial for applications like healthcare, smart homes and industrial automation.

Machine learning (ML) techniques have emerged as a practical and successful approach to this problem, enabling rapid detection and classification of various types of network threats [13]. Machine Learning allows the IDS to analyze past data to learn and optimize its performance. Additionally, ML-based computer algorithms may learn on their own with little to no human guidance. Supervised machine learning-based intrusion detection systems are most emerging due to their accurate prediction models [14,15]. Hybrid intrusion detection techniques employing machine learning approaches and deep learning techniques are popular for AIDS [16]. Although machine learning models still fall short in accurately identifying intrusions, the intrusion detection approach remains at the center of a wide variety of ongoing investigations. The major contributions of this study are as follows:

- To investigate the widespread usage of various datasets in Machine Learning-based Intrusion Detection Systems (IDS), this study analyzes 15 benchmark datasets, including ToN_IoT [17], CICIDS [18], KDD99 [19], UnswNB [20], IoT-BotNet [21], Wustl [22], Biflow-MQTT [23], BiFlow-DoS [24], UniflowDoS [25], CICDoS [26], CAIDA [27], NSL-KDD [28], DARPA [29], IoTNetIDS [30] and UHNIDS [31].
- We utilized five supervised and one semi-supervised machine learning techniques such as k-nearest neighbors (KNN), Random Forest (RF), Decision Tree (DT), Naive Bayes (NB), Artificial Neural Network (ANN) and

Auto-Encoder (AE) to train the models with both normal and attack parameters, presenting the results by evaluating key metrics: accuracy, recall, precision, F1-Score, ROC and execution time.

- A comprehensive comparison across different datasets is provided to understand the most influential factors.
- Additionally, a comparative analysis of results obtained from various machine learning approaches is presented through diverse types of charts.

Subsequent sections of the paper follow such organization: [Section 2](#) elaborates on background and related recent research, [Section 3](#) provides a detailed consideration of a range of datasets, while performance evaluation of all the 15 datasets is proposed along with experimental results, [Section 3](#). Finally, the paper is concluded in [Section 4](#).

2. Background and related work

Recent research indicates significant development and enhancement of Intrusion Detection Systems (IDS). In this section, we discuss some rigorous research conducted on IDS.

2.1. Supervised models for IDS framework

Several supervised anomaly detection techniques have been proposed in literature that can efficiently recognize anomalous behavior using labelled datasets. Altulaihan et al. [32] recently presented kNN-based DoS attack detection in IoT networks. This algorithm classifies the samples according to their similarity to labeled instances in the feature space. KNN is a very simple approach and it functions quite well when dealing with relatively small datasets, but it is rather computationally costly for large dataset size [33]. Bakro et al. [34] presented the RF-based ensemble technique combining multiple decision trees for enhanced robustness and accuracy. It reduces the chances of overfitting through the average prediction by individual trees as RF performs very well with high-dimensional data. RF minimizes overfitting by taking average predictions of individual trees as RF performs well on high-dimensional data [35]. RF minimizes overfitting [35] by averaging the predictions of individual trees and performs well on high-dimensional data. DT is used by Almseidin et al. [36] which offer interpretable models as data partitions occur recursively based on thresholds over features. Although efficient and interpretable, DTs become sensitive to complex datasets unless pruned or used in an ensemble like RF. NB classifiers employ probabilistic modeling according to Bayes' theorem, under the feature independence assumption. Interestingly, NB classifiers still perform surprisingly well for many scenarios, especially when computational efficiency and simplicity are the utmost concerns [37,38]. Cengiz et al. [39] presented ANN-based

approach for network intrusion detection. ANNs apply best in situations involving large-scale, high-dimensional datasets wherein they can automatically learn from complex patterns. Saheed et al. [40] introduced deep Long Short-Term Memory (LSTM) network to detect cyberattacks in IoT networks. LSTM model is fine tuned using a modified genetic algorithm. These supervised techniques constitute powerful tools for anomaly detection in intrusion systems, each with its strengths and limitations.

2.2. Unsupervised models for IDS framework

Unsupervised learning models play a very important role as they identify anomalies or patterns in data without requiring labeled samples, making them highly adaptable to dynamic and diverse environments. Srivastava et al. [41] presented an Entropy-Based Graph (EBG) model to classify network intrusions. This method discovers anomalous nodes or substructures of the graph, making it particularly effective for network-based intrusion detection. Another most common method is clustering-based methods such as K-Means which was used by Samunnisa et al. [42] for cloud intrusion detection. Principal component analysis (PCA) is another unsupervised technique to reduce dimensionality in a large-scale intrusion dataset [43]. It can be combined with supervised classification techniques. Zhang et al. [44] implemented a clustering algorithm called Density-Based Spatial Clustering of Applications with Noise (DBSCAN) in designing the IDS scheme for Wireless Sensor Networks. It is highly suitable for use in dealing with unlabeled and dynamic datasets.

2.3. Semi-supervised models for IDS framework

Semi-supervised learning models have become an important part of the Intrusion Detection Systems' framework. The model combines the strengths of supervised and unsupervised learning to detect anomalous or intruding activities in a system. For example, the application of an autoencoder in intrusion detection in internet of things networks was first introduced by Saheed et al. [45]. Autoencoders have also come up as highly useful components in semi-supervised IDS frameworks. Initially trained in an unsupervised mode to reconstruct input data, autoencoders learn compact and representative feature embeddings. Graph-based methods, such as Label Propagation and Graph Convolutional Networks (GCNs), have been very promising within the framework of IDS by using graph structures to propagate labels across unlabeled instances. Hybrid approaches include the semi-supervised GANs [46], in which generative and discriminative capabilities are combined with refinements of decision boundaries involving mixed real and pseudo-labeled data.

2.4. Hybrid models for IDS framework

Hybrid models of Intrusion Detection Systems (IDS) frameworks combine multiple techniques in order to use the strengths of various approaches. Thus, they improve the accuracy and adaptability. In such hybrid models, signature-based detection of known threats can often be combined with anomaly-based detection for novel attacks. For example, using unsupervised methods such as clustering in combination with Decision Trees machine learning models would improve detection capabilities on both labeled and unlabeled data. Furthermore, hybrid deep learning frameworks can be built by combining feature extraction from autoencoders with supervised classifiers such as Random Forests to solve robust solutions for high-dimensional complex datasets. Arthi et al. [47] proposed a hybrid approach combining machine learning and deep learning techniques in the form of DNN and SVM to detect network intrusion in sensor-based healthcare data. Jan et al. [48] provided the comparative study of all kinds of machine learning algorithms with regard to intrusion detection systems, which includes supervised, unsupervised, reinforcement learning and deep learning. They also highlighted the concept of data visualization in designing the intrusion detection system. Another study presented a hybrid model that combines a fused CNN with a bidirectional gated recurrent unit (Bi-GRU) for the detection and classification of intruders. The Bi-GRU model is equipped with an attention mechanism to identify the most important features responsible for DDoS attack detection [49]. From the above literature, we have identified several important research aspects: Supervised anomaly detection techniques that include kNN, RF, DT, NB and ANNs have been extensively used to analyze intrusion detection using labeled datasets. Our paper tests these models across different IDS datasets. Semi-supervised techniques, like Autoencoder (AE), bridge the gap between supervised and unsupervised learning [50]. This work examines the feasibility of such technique for improving the performance of an IDS. Hence, we analyze the performance of various supervised and semi-supervised algorithms with 15 datasets related to IoT. This analysis can be beneficial to understand the performance of the algorithms over large and small datasets.

3. Datasets

This work analyzes 15 different datasets. They range from a wide variety of attack types, including DoS, botnet and intrusion attempts, which allows for a complete evaluation of the solution in different threat scenarios. They also represent various network environments, such as IoT, like TON_IoT IoT-BotNet, industrial IoT, like Wustl and general network traffic, like CICIDS, KDD99, making the results more realistic. Most of the datasets are already established in the research community, such as CICIDS, NSLKDD and CAIDA,

where comparisons can be made with prior studies and validate results. Moreover, the scale and complexity of the datasets vary from small datasets such as KDD99 to larger and more complex ones like CICIDS and IoTNetIDS, which provides insight into how machine learning models perform on both small and large scales. This aggregation of factors helps ensure that the datasets selected present IDS models with full and representative assessments.

3.1. KDD99

KDD99 makes use of the same labels of DARPA and has four kinds of features, namely: basic features, content features, host-based statistical features and time-based statistical features. Unfortunately, the KDD99 dataset is fraught with flaws. The dataset is highly imbalanced, which means the classification results are biased to the dominant classes. Further, the database has a lot of duplicates and redundancies; therefore, several researchers filter many of the records before applying them in experiments. Hence, sometimes the experimental results from different studies cannot be compared directly. Finally, the KDD data is dated, which means it does not serve well in describing the network environment of the current era. [Table 1](#) presents the attack taxonomy.

3.2. NSL_KDD

NSL-KDD was constructed to address the inadequacies of the KDD99 dataset. Records in the NSL-KDD were thoughtfully selected with guidance from KDD99. NSL-KDD balances records across multiple classes, mitigating the classification bias problem. Furthermore, duplicate and redundant records were removed, resulting in a manageable quantity of records. Therefore, experiments can be performed on the whole data set with the advantage of comparable and consistent results of various experiments. This dataset partly addresses issues such as data bias and redundancy. However, it lacks any new data, resulting in the absence of samples from the minority class and out-of-date data. [Table 2](#) describes the attack taxonomy.

Table 1. Attack taxonomy of KDD99 dataset.

Attacks	Normal	DOS	R2L	U2R	Probe	Total
Data Sample	67343	45927	11656	995	52	125973

Table 2. Attack taxonomy of NSL_KDD dataset.

Attacks	Normal	DOS	R2L	U2R	Probe	Total
Data Sample	391458	97277	1126	52	4107	494020

3.3. UNSW-NB15

This dataset was created by the University of New South Wales, which used a tool called Bro to extract 49-dimensional features from network traffic captured by three virtual servers. It includes more types of attacks than the KDD99 dataset and more features. Data categories include normal data and nine forms of attacks. Features in the dataset encompass flow features, basic features, content features, time features, additional features and labeled features. UNSW-NB15 represents a novel IDS dataset that has been employed in numerous recent investigations. Although it currently holds less influence than KDD99, the creation of new datasets is imperative for the development of new IDS based on machine learning. Table 3 provides an overview of the attack taxonomy.

3.4. DARPA99

DARPA99 is the dataset developed by the Lincoln Laboratory of MIT that acts as a benchmark frequently used in IDS research. The dataset compiled for 9 weeks has 7 weeks for training and the remaining 2 weeks as a test set. Both raw packets and their labels are included in this dataset. The labels include five types: Normal, Denial of Service (DOS), Probe, User to Root (U2R) and Remote to Local (R2L). The KDD99 dataset was created to address the challenge of applying raw packets directly to conventional machine learning models. Table 4 presents the attack taxonomy.

3.5. ToN_IoT

Different data samples retrieved from Telemetry datasets of IoT and IoT devices have been consolidated into datasets labeled ToN IoT. The data are retrieved from the Cyber Range and IoT Labs at UNSW Canberra's School of Engineering and Information Technology (SEIT) which established a realistic large-scale network, including Industrial 4.0, IoT and IoT networks as a new testbed. Different types of attacks, including DoS, DDoS and ransomware, are performed throughout the IoT network, targeting web applications, IoT gateways and computer systems. The datasets were gathered using parallel processing to capture

Table 3. Attack taxonomy of UNSW-NB15 dataset.

Attacks	Normal	DDOS	DoS	Injection	Shellcode	Backdoor	Total
Data Sample	56000	10000	1746	12264	5000	1000	86010

Table 4. Attack taxonomy of DARPA99 dataset.

Attacks	Normal	DDOS	R2L	U2R	Total
Data Sample	86795	44362	13897	7456	152600

Table 5. Attack taxonomy of ToN_IoT dataset.

Attacks	Normal	DDOS	XSS	Injection	Backdoor	Ransomware	Worm	Total
Data Sample	35000	5000	2042	5000	3000	5000	820	55862

a variety of normal and cyber-attack events from network traffic, Windows audit traces, Linux audit traces and IoT service telemetry data. Table 5 gives an overview of the attack taxonomy.

3.6. IoT_BOTNET

This dataset was created in the UNSW Canberra Cyber Range Lab, simulating a realistic network environment with a blend of normal and botnet traffic. The source files for the dataset are accessible in various formats, including original pcap files, generated argus files and csv files. To facilitate the labeling process, the files were categorized by attack type and subcategory. The pcap files have a size of 69.3 GB and consist of approximately 72,000,000 records. Table 6 illustrates the attack taxonomy.

3.7. IoTNetIDS

This dataset was derived from the IoT environment of a smart house. The smart house consists of AI speakers (SKTNGU), Wi-Fi cameras (EZVIZ), computers, smartphones, tablets, a wireless access point (Wi-Fi) and a Wi-Fi router, among other networked components. Cameras and AI speakers were used to simulate IoT victim equipment, while other devices represented attacking devices. The testbed was designed using the Network Mapper tool to mimic a host of real-world attacks within the IoT ecosystem. Table 7 shows a summary of the attack taxonomy.

3.8. CICIDS and CICDoS

CICIDS and CICDoS datasets offer both benign and contemporary common attacks, simulating authentic real-world data in the form of PCAPs. They also include the outcomes of network traffic analysis using CICFlowMeter, which

Table 6. Attack taxonomy of IOT_BOTNET dataset.

Attacks	Normal	DDOS	XSS	Injection	Ransomware	Botnet	Total
Data Sample	90000	25000	2042	10000	9830	12565	149437

Table 7. Attack taxonomy of IoTNetIDS dataset.

Attacks	Normal	DDOS	Eavesdropping	Worm	Total
Data Sample	119000	33000	1500	12000	203518

Table 8. Attack taxonomy of CICIDS dataset.

Attacks	Normal	DDOS	DOS	Heartbleed	Botnet	Brute force	Total
Data Sample	15000	2700	3000	5000	1100	1100	27900

Table 9. Attack taxonomy of CICDOS dataset.

Attacks	Normal	DDOS	DOS	Injection	Total
Data Sample	17856	7564	8600	3368	37388

consists of labeled flows based on timestamp, source and destination IP addresses, source and destination ports, protocols and attack vectors presented in CSV files. These datasets cover DDOS and DOS attacks. [Tables 8 and 9](#) depict the attack taxonomy for the CICIDS and CICDOS datasets, respectively.

3.9. CAIDA

This collection incorporates anonymized passive traffic traces from several CAIDA high-speed monitors on high-speed Internet backbone links. CryptoPan prefix-preserving anonymization is employed to anonymize the traffic traces in this dataset. The anonymization key varies annually and is consistent for all traces collected within the same calendar year. To prevent excessive packet loss caused by disk I/O overload, packets are truncated at a snap length specified during capture. Historically, the snap length has ranged from 64 to 96 bytes. Additionally, all packets' payloads are removed, retaining only header information up to layer 4 (transport layer). [Table 10](#) provides an overview of the attack taxonomy.

3.10. WUSTLE-IIoT

WUSTL-IIoT is a network data collection system for the Internet of Things (IoT), designed for use in cybersecurity research. The dataset was generated utilizing a specified IoT testbed. The objective of our testbed is to accurately simulate real-world industrial systems while allowing for realistic cyber-attacks. Over the course of 53 h, we have collected 2.7 GB of data. The dataset has undergone pre-processing and sanitization, which includes eliminating missing values, corrupted values (i.e. invalid entries) and severe outliers. We have used a reduced version of the dataset, slightly over 400 MB, and provided it here. [Table 11](#) presents the attack taxonomy.

Table 10. Attack taxonomy of CAIDA dataset.

Attacks	Normal	DDOS	DOS	Worm	Backdoor	Total
Data Sample	36000	12000	2275	4772	9983	65030

Table 11. Attack taxonomy of WUSTLE dataset.

Attacks	Normal	DDOS	XSS	Botnet	Injection	Infiltration	Total
Data Sample	100000	5000	2042	54265	3000	7783	172090

3.11. Biflow_MQTT and Biflow_DOS

The dataset was specifically designed to represent particular networks or protocols, such as IoT-based networks and Supervisory Control and Data Acquisition (SCADA) networks. Specialized networks like these incorporate unique devices not commonly found in other network types. These devices are purpose-built for such networks, such as remote terminal units in SCADA networks and others. These datasets encompass DOS and brute force attacks. [Tables 12 and 13](#) present the attack taxonomy for the Biflow_MQTT and Biflow_DOS datasets, respectively.

3.12. UHNIDS

The Unified Host and Network Dataset is a subset of network and computer (host) events collected over a 90-day period from the Los Alamos National Laboratory enterprise network. To preserve the security of LANL's operational IT environment, the data values have been anonymized. Both host and network data have matching identities, enabling their joint use for analysis and research. Values were not anonymized in specific circumstances, including well-known network ports, system-level usernames (not associated with individuals) and system-level hosts. In some instances, hosts representing well-known redundant services, such as Active Directory servers, LANL's email servers and LANL's automatic vulnerability detection systems, were included. [Table 14](#) provides an overview of the attack taxonomy.

Table 12. Attack taxonomy of Biflow_mqtt dataset.

Attacks	Normal	DDOS	Bruteforce	Total
Data Sample	6523	2152	9844	18519

Table 13. Attack taxonomy of Biflow_dos dataset.

Attacks	Normal	DDOS	Total
Data Sample	49625	45513	95138

Table 14. Attack taxonomy of UnifiedHNIDS dataset.

Attacks	Normal	DDOS	Fuzzers	Exploits	Total
Data Sample	23226	15700	3866	1770	44562

3.13. UniflowDoS

The dataset was specifically designed with a particular objective: to represent specific networks or protocols, such as IoT-based networks and Supervisory Control and Data Acquisition (SCADA) networks. These specialized networks include various unique devices not commonly found in other network types. These devices are purpose-built for such networks, such as remote terminal units in SCADA networks, among others. The datasets encompass DOS and brute force attacks. Table 15 presents the attack taxonomy.

This data sampling is aimed at understanding the types of attacks available in each dataset and the total number of data samples. Getting a full view of the entire scenario helps to be very useful in data pre-processing. Different optimization techniques can be used depending on the number of data samples. Data samples are large in the IoT-BotNet, Wustl and IoTNetIDS datasets. Therefore, these datasets require normalization and standardization, especially to enable machine learning models to perform quickly and enable training with high accuracy. Table 16 summarizes all the 15 datasets as a whole.

Figure 2 graphically illustrates the attack samples from all datasets. This will make it easy to realize quickly and intuitively both the proportion and spread of attack instances in each data set. From the graph description, one can derive the kinds of patterns, differences and relative frequencies of different types of attacks, which will, in turn, provide information on the characteristics of distribution of attacks in each data set. Such visualizations are of great helpfulness while identifying trends, anomalies or concentrations of some attacks and thus aid in the research and analysis on the dataset's attack landscape.

Table 15. Attack taxonomy of UniflowDoS dataset.

Attacks	Normal	DDoS	Bruteforce	Total
Data Sample	9730	7756	5690	23176

Table 16. Comparison of all datasets.

Datasets	Types of attacks	Label data	Data metric	Year
DARPA [29]	DOS, R2L, U2L, DDOS	YES	Network	1999
KDD99 [19]	DOS, R2L, U2L	YES	Network	1999
NSL-KDD [28]	DDOS, DOS, probe	YES	Biometric, Network	2009
UnswNB [20]	Backdoors, DDOS, shellcode, worms, DOS	YES	Network	2015
Biflow-MQTT [23]	Brute force, DDoS	NO	Network	2016
CICIDS [18]	Brute force, DDOS, Heartbleed, Botnet	YES	Network	2017
ToN_IoT [17]	DOS, DDOS, ransomware, worms, XSS,	YES	Network	2017
CICDoS [26]	DOS, DDOS	YES	Network	2017
IoT-BotNet [21]	DDOS, Botnet, XSS, Injection	YES	Biometric, Network	2018
UHNIDS [31]	Fuzzers, DOS, exploits, worms	YES	Biometric, Network	2018
CAIDA [27]	DDOS, worm, backdoor	YES	Network	2019
Wustl [22]	Botnet, infiltration, injection, DDOS, XSS	YES	Biometric, Network	2020
BiFlow-DoS [24]	DDOS, DOS	NO	Network	2020
UniflowDoS [25]	Brute force, DDOS	NO	Network	2020
IoTNetIDS [30]	DDOS, Eavesdropping, worms	NO	Biometric, Network	2020

4. Performance evaluation

To evaluate the above datasets, we used five different kinds of supervised machine and semi-supervised learning algorithms: K-Nearest Neighbour, Support Vector Machine, Decision Tree, Random Forest, Artificial Neural Network and Auto-Encoder. We have utilized the Pandas Library and Scikit-Learn Library in Python to preprocess, train and test the dataset. First, we

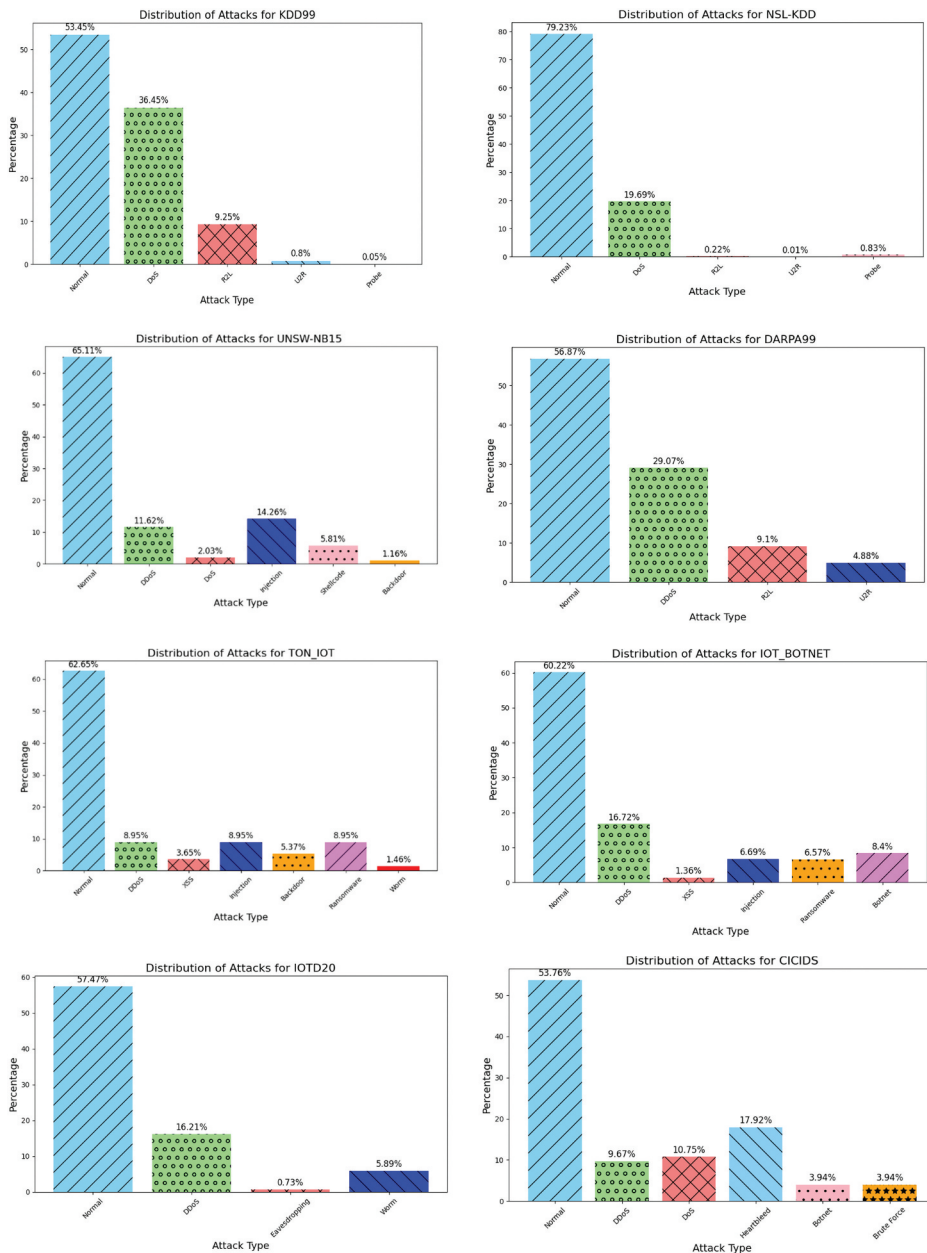


Figure 2a.

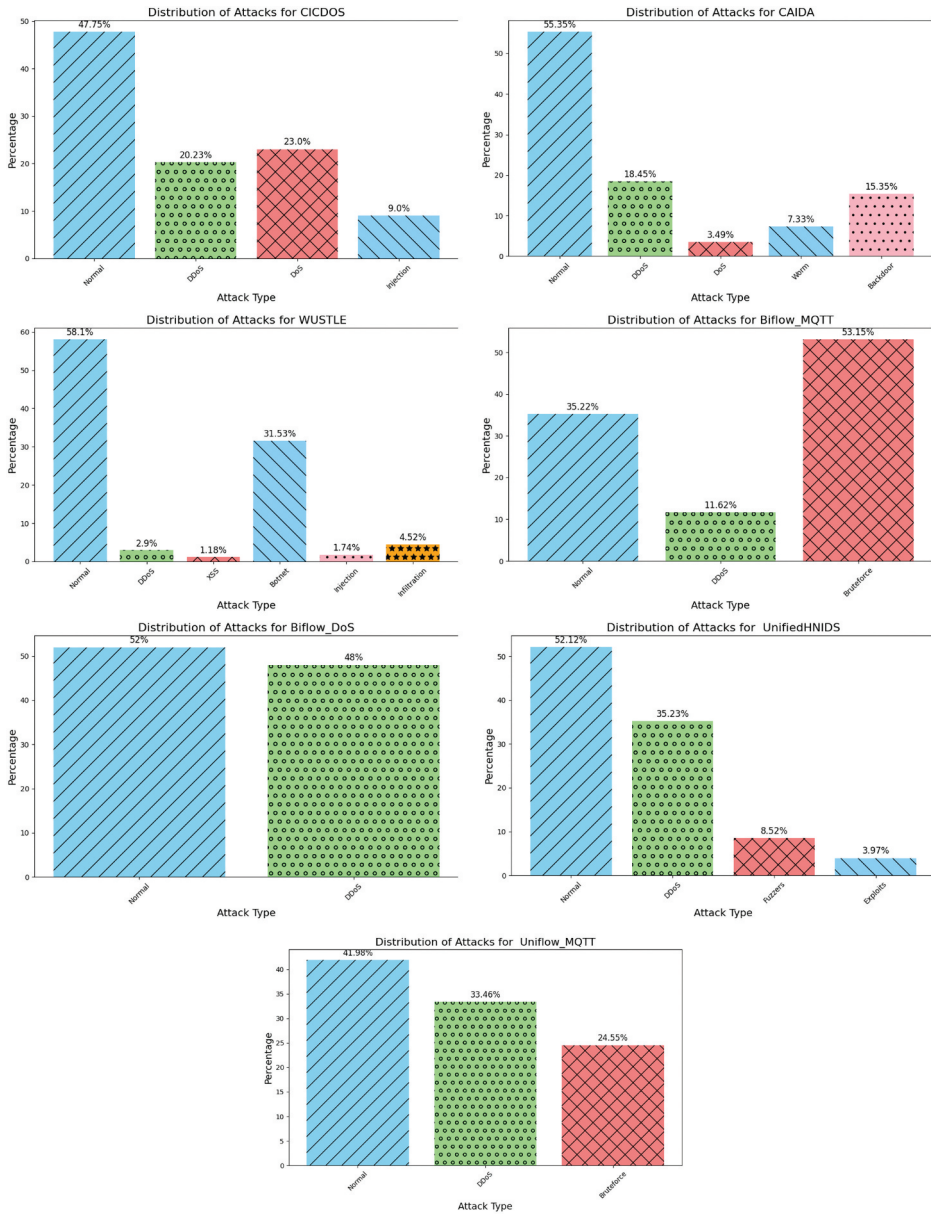


Figure 2b. Attack taxonomy of all datasets.

imported the above datasets and necessary libraries in which we extracted both the dependent and independent variables. After that, encoding was done to convert categorical data into numerical values. Then, Scikit-Learn was used to split the dataset into training and testing sets, maintaining a ratio of 75% to 25%. In some cases, even after encoding, the dataset might be large enough and could create problems. So, we applied feature scaling to keep all the data within a particular range. This procedure, known as standardization, was carried out

through the use of the Standard Scalar library from Scikit-Learn. Next, the models were trained, and once finished, we then tested the models with multiple performance metrics: accuracy, recall, precision, F1-Score, ROC and execution time. We applied the same set of experiments for k-fold cross-validation also. The details for each metric follow:

True Positive (*TP*):- Occurs when an actual positive instance is correctly predicted as positive.

True Negative (*TN*):- Occurs when an actual negative instance is correctly predicted as negative.

False Positive (*FP*):- Occurs when an actual negative instance is incorrectly predicted as positive.

False Negative (*FN*):- Occurs when an actual positive instance is incorrectly predicted as negative.

Accuracy: This metric gauges the overall success of the model by comparing the number of correctly classified attack and normal instances to the total number of instances.

$$\frac{TP + TN}{(TP + TN + FP + FN)}$$

Precision: This metric assesses the overall effectiveness of the model by determining the percentage of observations identified as attacks that are indeed attack observations.

$$\frac{TP}{TP + FP}$$

Recall: This metric measures the overall success of the model by determining the percentage of actual attack observations that are correctly classified.

$$\frac{TP}{TP + FN}$$

F1-Score: Calculated as the harmonic mean of precision and recall, the F1-Score is a statistical function used to evaluate the accuracy of a model. As precision and recall approach 100%, the F1-Score and accuracy reach their maximum, ensuring that every instance is classified correctly.

$$\frac{2 \times Precision \times Recall}{Precision + Recall}$$

Execution Time: This denotes the time taken by all the models to make predictions differentiating between normal and attack data.

5. Results and analysis

We utilized 15 real traffic and device metric IoT datasets to train our machine learning models, including KNN, RF, DT, NB, ANN and AE. We applied both hold-out and 10-fold methods to all datasets for all algorithms. The outcome of these experiments is significantly influenced by the dataset, making it crucial to comprehend the imbalanced data as it affects both accuracy and precision. The focus of this analysis is to determine if various machine learning models exhibit a better performance rate under different data metrics in unbalanced datasets. Table 17 presents the accuracy and precision for the 15 datasets for the hold-out method. Here, we split the dataset in 75:25 training and testing ratio.

Performance analysis on different datasets shows that algorithms are quite different in various aspects. kNN performed consistently with high accuracy and precision throughout all the datasets, like ToN_IoT and UniflowDoS, but had slightly lower reliability at KDD99. Random Forest (RF) also gave promising results, especially with IoT-BotNet and UniflowDoS, but the precision was much lower than the accuracy in this case, as in KDD99. The DT gets high accuracy in IoTNetIDS and IoT-BotNet datasets but failed on KDD99. Hence, this might be the data-imbalance sensitivity. The NB is fairly accurate on a balanced dataset such as IoTNetIDS but performs poorly on precision and accuracy with greater complexity such as CICIDS. ANN exhibits good performance on all the datasets, with excellent accuracy in IoTNetIDS and UniflowDoS but with slightly less precision on KDD99. Finally, Auto-Encoder obtains good accuracy and precision for structured datasets such as Wustl and UniflowDoS with high adaptability but falters with slightly less precision on NSL-KDD. In general, ANN and kNN appear to be safe algorithms, whereas RF and Auto-Encoder have high precision in particular cases.

Table 17. Accuracy and precision comparison across all models and datasets for hold-out method.

Dataset	kNN		RF		DT		NB		ANN		Encoder	
	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc
ToN_IoT	0.98	0.97	0.89	0.88	0.77	0.82	0.53	0.65	0.95	0.97	0.95	0.96
Biflow-MQTT	1.00	0.98	0.96	0.97	1.00	0.97	1.00	0.97	0.99	0.99	0.94	0.94
Wustl	0.93	0.94	0.93	0.93	0.89	0.92	0.93	0.86	0.92	0.96	1.00	0.99
UnswNB	0.99	0.98	0.98	0.95	0.99	0.96	0.37	0.66	0.98	0.96	0.95	0.95
IoT-BotNet	0.99	0.99	0.98	0.94	1.00	0.97	1.00	0.98	0.98	0.98	0.91	0.91
IoTNetIDS	1.00	1.00	0.97	0.97	0.99	0.98	0.99	0.80	0.99	0.99	0.92	0.93
KDD99	0.90	0.90	1.00	0.80	0.69	0.75	0.18	0.28	0.90	0.88	0.87	0.91
CICIDS	0.98	0.98	0.88	0.87	0.88	0.88	0.98	0.36	0.88	0.89	0.91	0.92
DARPA	0.97	0.94	0.98	0.93	0.88	0.93	0.99	0.82	0.99	0.92	0.97	0.96
CAIDA	0.97	0.95	0.90	0.93	0.88	0.93	0.84	0.94	0.99	0.97	0.90	0.90
CICDoS	0.98	0.98	0.88	0.87	0.88	0.88	0.98	0.36	0.88	0.89	0.93	0.96
UniflowDoS	1.00	0.98	0.96	0.97	1.00	0.97	1.00	0.97	0.99	0.99	0.99	0.98
UHNIDS	0.97	0.95	0.90	0.93	0.88	0.93	0.84	0.94	0.99	0.97	0.91	0.89
NSL-KDD	0.99	0.97	1.00	0.82	0.90	0.93	0.66	0.78	0.90	0.94	0.86	0.84
BiFlowDoS	1.00	0.98	0.96	0.97	1.00	0.97	1.00	0.97	0.99	0.99	0.98	0.99

Acc= Accuracy, Prc=Precision.

Table 18. Recall and F1 score comparison across all models and datasets for hold-out method.

Dataset	kNN		RF		DT		NB		ANN		Encoder	
	RC	F1	RC	F1	RC	F1	RC	F1	RC	F1	RC	F1
ToN_IoT	0.98	0.97	0.76	0.81	0.70	0.73	0.77	0.78	0.97	0.96	0.88	0.87
Biflow-MQTT	0.97	0.98	0.98	0.97	0.95	0.98	0.97	0.98	1.00	1.00	0.98	0.93
Wustl	0.98	0.81	0.99	0.63	0.98	0.62	0.49	0.46	0.97	0.95	0.90	0.93
UnswNB	0.96	0.99	0.75	0.85	0.80	0.89	0.95	0.53	0.95	0.97	0.88	0.87
IoT-BotNet	0.99	0.99	0.75	0.85	1.00	0.93	0.99	0.99	0.97	0.88	0.82	0.99
IoTNetIDS	1.00	1.00	1.00	0.98	1.00	0.99	0.80	0.88	1.00	0.99	0.94	0.99
KDD99	0.68	0.99	0.55	0.36	0.94	0.66	1.00	0.31	0.70	0.80	0.86	0.88
CICIDS	0.99	0.98	1.00	0.94	1.00	0.92	0.69	0.16	1.00	0.94	0.91	0.91
DARPA	0.90	0.3	0.88	0.96	0.97	0.96	0.78	0.88	0.91	0.89	0.92	0.93
CAIDA	0.88	0.92	0.97	0.92	0.99	0.91	0.95	0.93	0.95	0.88	0.87	0.91
CICDoS	0.99	0.98	1.00	0.94	1.00	0.92	0.69	0.16	1.00	0.94	0.97	0.95
UniflowDoS	0.97	0.98	0.98	0.97	0.95	0.98	0.97	0.98	1.00	1.00	0.92	0.99
UHNIDS	0.88	0.92	0.97	0.92	0.99	0.91	0.95	0.93	0.95	0.88	0.99	0.99
NSL-KDD	0.86	0.99	0.85	0.86	0.94	0.86	1.00	0.61	0.90	0.94	0.92	0.92
BiFlowDoS	0.97	0.98	0.98	0.97	0.95	0.98	0.97	0.98	1.00	1.00	0.89	0.98

RC= Recall, F1=F1 Score.

The recall (RC) and F1 scores evaluation over different algorithms and datasets shows different trends in performance, as shown in [Table 18](#). kNN has high recall and F1 performance over most of the datasets, especially IoTNetIDS and Biflow-MQTT, but low recall over KDD99. Random Forest (RF) has competitive recall performances in datasets like UniflowDoS and CICDoS, but its F1 score is considerably lower in Wustl and KDD99, indicating a challenge in dealing with imbalanced data. Decision Tree (DT) gets high recalls in IoTNetIDS and CICDoS, but the F1 score drops on Wustl and KDD99. It is sensitive to data complexity. Naive Bayes has a good recall on the datasets NSL-KDD and IoT-BotNet, but its F1 score is significantly worse on CICIDS and DARPA. It struggled with precision. ANN produces strong recall and F1 scores for nearly all datasets, while significantly outperforming the rest in UniflowDoS and BiFlowDoS, which means that the model is quite adaptable. Finally, Auto-Encoder produces high F1 scores in structured datasets such as IoTNetIDS and UHNIDS, while it does not perform that great on CAIDA and KDD99. Overall, ANN and kNN are reliable models, while RF and Auto-Encoder show strengths in particular contexts but struggle with highly imbalanced or complex datasets. [Figures 3 and 4](#) show the accuracy and Precision Comparison and ROC comparison for all algorithms, respectively.

[Figure 4](#) shows the performance of the algorithms for different datasets. The greater Area Under Curve (AUC) depicts more discrimination between classes. From the curves, ANN and kNN algorithms were consistently able to show excellent AUC scores, thus indicating their stability in detecting anomalies. Naive Bayes (NB) type algorithms have AUC scores in some of the datasets low, which also indicates a lack of capability on their part in dealing with complex patterns. In summary, the ROC analysis puts ANN and kNN in the top ranks, while Auto-encoder and RF perform very well in some datasets.

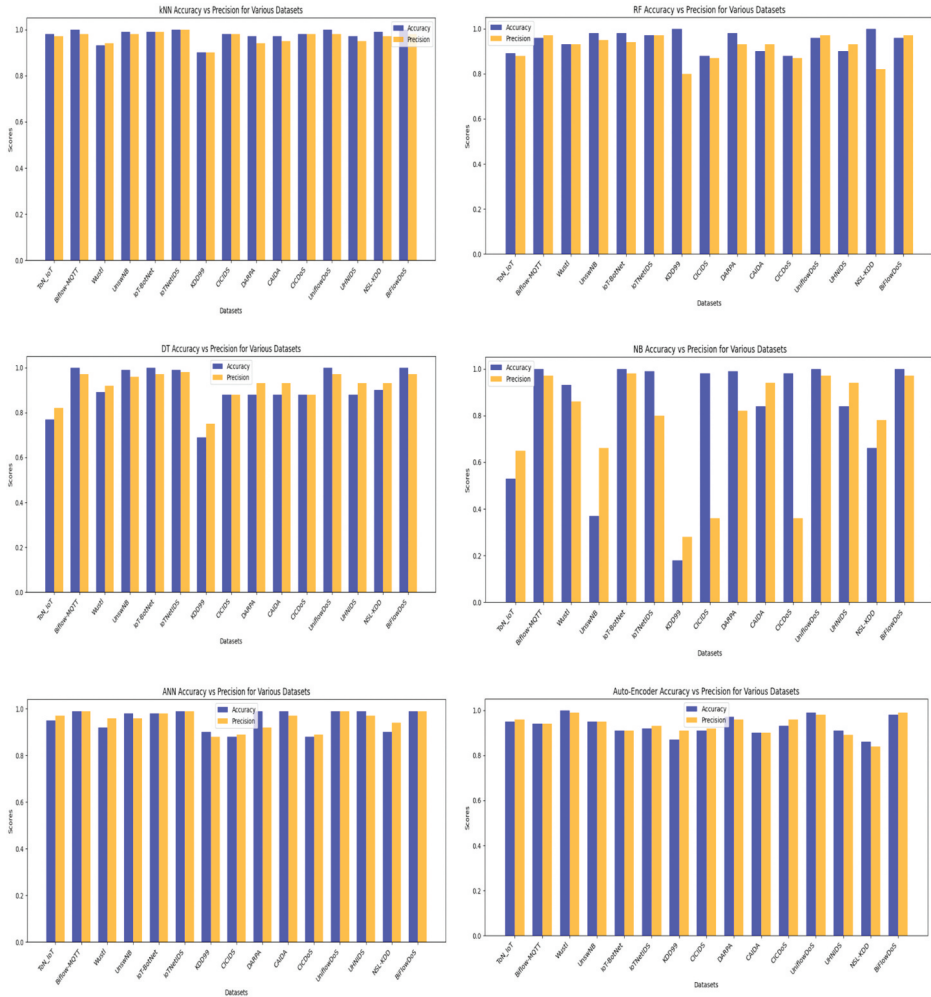


Figure 3. Accuracy and precision comparison for all datasets for hold-out method.

Execution time comparison of different machine learning algorithms is shown in [Figure 5](#) which reveals notable patterns on training and test sets. NB is the most efficient at performing both training and test and, therefore, recorded less execution time on all data-sets. DT maintains moderate execution times, and, on the other hand, RF and ANN show higher execution time, especially while training, since these have more complex structure and design. KNN is time-consuming when testing, particularly for larger datasets, since it calculates distances. AE has balanced performance with average times in both training and testing. Overall, NB and DT have the best performance concerning efficiency, while ANN and RF are resource hungry.

Similarly, we experimented k-fold method with all algorithms and datasets. [Tables 19 and 20](#) show the Accuracy, Precision, Recall and F1 Score across all algorithms and datasets for K-fold (10-fold) cross validation approach. It can be

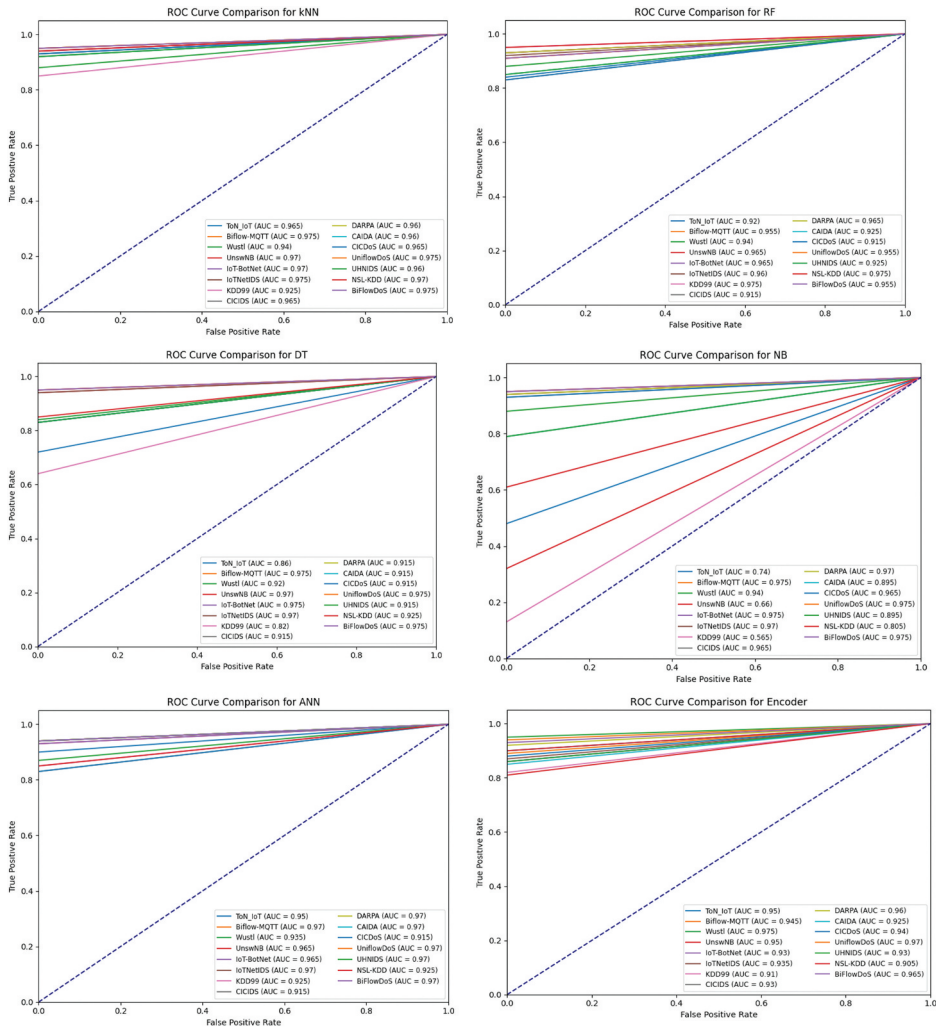


Figure 4. ROC comparison for all datasets and algorithms for hold-out method.

noted from Figure 6 that kNN's performance across datasets using the k-fold cross-validation method is consistently high, especially for datasets such as IoTNetIDS and Biflow-MQTT, where kNN attained an accuracy of 1.00 and a precision of 0.99, which reflects the robustness and reliability of these datasets. However, it drops down a little with datasets KDD99 and CAIDA, wherein kNN achieved an accuracy of 0.89 and 0.95, respectively. This implies that it is not perhaps as effective in handling certain data distributions. Random Forest (RF) performs well on k-fold validation, where accuracy values are in the range of 0.93–0.95 for datasets such as Wustl and UnswNB. The precision values are mainly high, which means that the model is making good class distinctions. For simple datasets such as ToN_IoT, the accuracy improves with RF compared to the hold-out method, as it is 0.88, which means RF benefits from the iterative

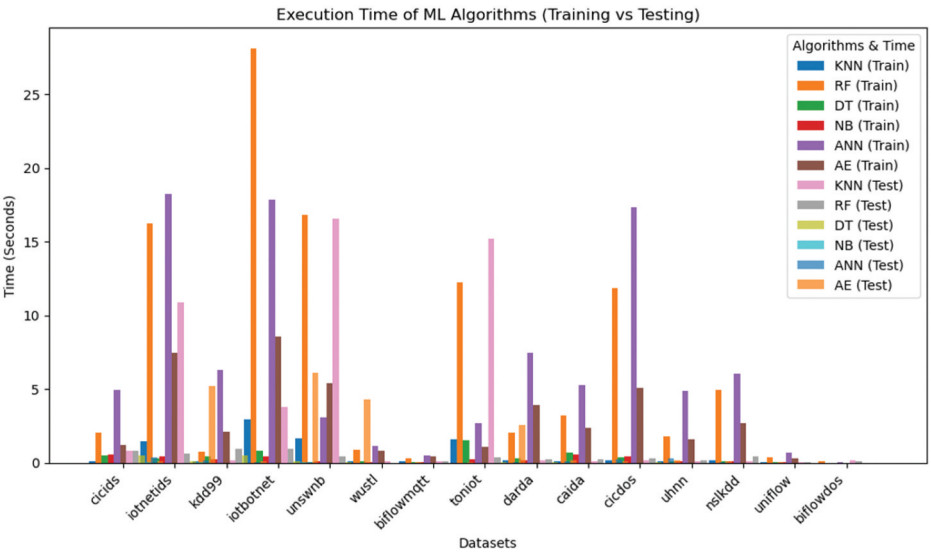


Figure 5. Training and testing time comparison for all datasets and algorithms for hold-out method.

Table 19. Recall and F1 score comparisons across all models and datasets for k-fold method.

Dataset	kNN		RF		DT		NB		ANN		Encoder	
	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc	Acc	Prc
ToN_IoT	0.96	0.97	0.88	0.90	0.77	0.81	0.54	0.62	0.95	0.96	0.94	0.95
Biflow-MQTT	1.00	0.99	0.97	0.96	0.98	0.95	0.99	0.96	0.99	0.98	0.93	0.94
Wustl	0.92	0.93	0.94	0.91	0.89	0.91	0.90	0.83	0.91	0.94	0.99	0.98
UnswNB	0.98	0.97	0.95	0.93	0.96	0.94	0.38	0.67	0.97	0.94	0.94	0.92
IoT-BotNet	0.98	0.99	0.96	0.92	0.99	0.95	0.98	0.95	0.97	0.98	0.89	0.90
IoTNetIDS	1.00	1.00	0.96	0.95	0.98	0.96	0.96	0.79	0.98	0.97	0.91	0.92
KDD99	0.89	0.88	0.98	0.79	0.67	0.74	0.17	0.26	0.88	0.87	0.85	0.90
CICIDS	0.97	0.96	0.87	0.85	0.87	0.86	0.96	0.35	0.87	0.88	0.89	0.90
DARPA	0.96	0.93	0.97	0.92	0.87	0.92	0.98	0.81	0.98	0.91	0.96	0.94
CAIDA	0.95	0.94	0.89	0.91	0.87	0.91	0.83	0.93	0.97	0.96	0.89	0.88
CICDoS	0.97	0.97	0.87	0.85	0.87	0.86	0.96	0.35	0.87	0.88	0.92	0.95
UniflowDoS	1.00	0.98	0.97	0.96	0.98	0.94	0.99	0.94	0.98	0.98	0.98	0.97
UHNIDS	0.96	0.95	0.89	0.92	0.86	0.92	0.82	0.93	0.98	0.95	0.90	0.89
NSL-KDD	0.98	0.96	0.99	0.81	0.89	0.92	0.64	0.77	0.88	0.93	0.85	0.83
BiFlowDoS	1.00	0.99	0.97	0.96	0.98	0.94	0.98	0.94	0.99	0.98	0.96	0.97

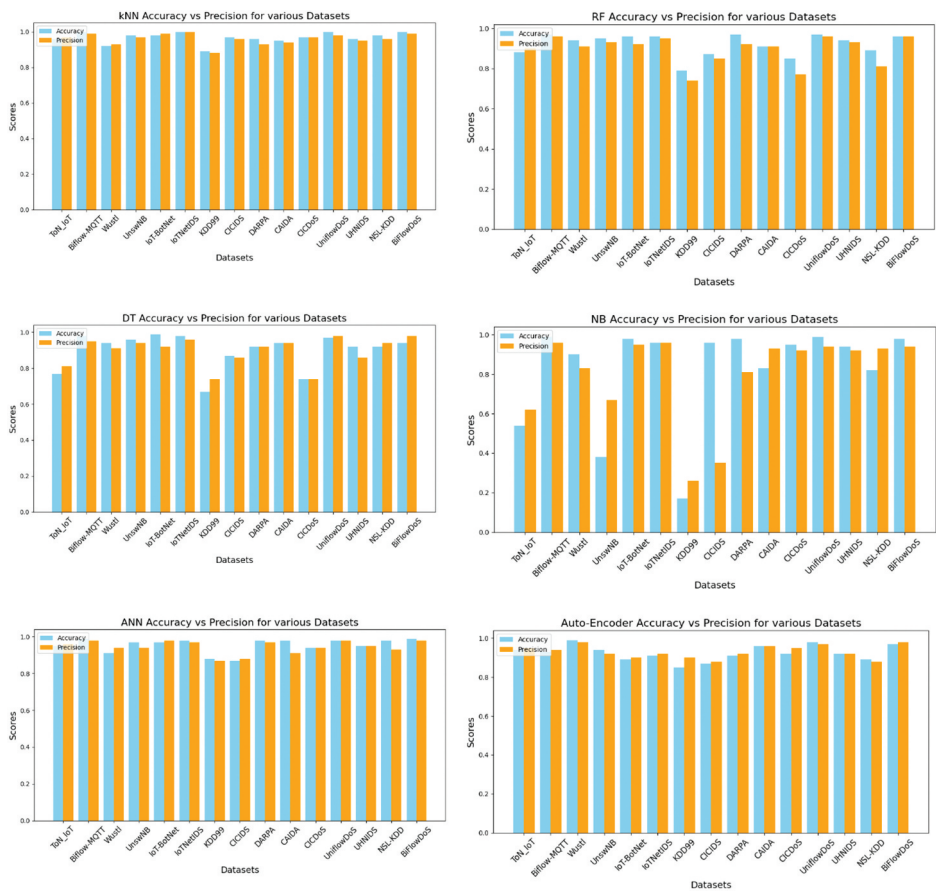
Acc= Accuracy, Prc=Precision.

evaluation of k-fold cross-validation. DT performs moderately under the k-fold method. For datasets such as IoT-BotNet and IoTNetIDS, it has a good performance, with accuracy values reaching nearly 0.98. However, for the complex datasets such as KDD99 and CICIDS, its accuracy drops drastically to 0.67 and 0.87, respectively. This variation indicates the fact that though DT supports structured data, it may fail for complex patterns. ANN performs excellently well on many datasets in k-fold validations, achieving very good performances on datasets such as Biflow-MQTT and IoTNetIDS, such that accuracy and precision attain 0.99 and 0.98, respectively. Encoder-based models show stable

Table 20. Recall and F1 score comparison across all models and datasets for k-fold method.

Dataset	kNN		RF		DT		NB		ANN		Encoder	
	RC	F1	RC	F1	RC	F1	RC	F1	RC	F1	RC	F1
ToN_IoT	0.97	0.96	0.79	0.85	0.72	0.76	0.75	0.80	0.96	0.95	0.90	0.89
BiFlow-MQTT	0.98	0.98	0.96	0.95	0.94	0.96	0.95	0.97	1.00	1.00	0.94	0.95
Wustl	0.94	0.83	0.93	0.65	0.92	0.68	0.50	0.47	0.95	0.94	0.89	0.91
UnswNB	0.97	0.98	0.77	0.82	0.81	0.88	0.94	0.55	0.93	0.95	0.91	0.88
IoT-BotNet	0.99	0.99	0.79	0.84	0.96	0.92	0.96	0.97	0.96	0.97	0.85	0.94
IoTNetIDS	1.00	1.00	0.98	0.96	0.98	0.95	0.83	0.89	0.99	0.98	0.93	0.97
KDD99	0.69	0.95	0.60	0.42	0.92	0.72	0.98	0.40	0.78	0.84	0.87	0.90
CICIDS	0.98	0.98	0.97	0.92	0.96	0.90	0.73	0.20	0.97	0.93	0.90	0.92
DARPA	0.91	0.35	0.91	0.95	0.96	0.94	0.82	0.87	0.90	0.88	0.89	0.92
CAIDA	0.87	0.90	0.96	0.91	0.98	0.93	0.93	0.90	0.95	0.89	0.88	0.91
CICDoS	0.98	0.97	0.96	0.94	0.98	0.93	0.73	0.21	0.98	0.95	0.91	0.92
UniflowDoS	0.97	0.98	0.99	0.96	0.97	0.95	0.96	0.96	1.00	1.00	0.94	0.96
UHNIDS	0.87	0.91	0.95	0.93	0.97	0.93	0.94	0.91	0.93	0.90	0.94	0.93
NSL-KDD	0.85	0.95	0.83	0.82	0.91	0.82	0.95	0.65	0.85	0.91	0.91	0.90
BiFlowDoS	0.99	0.99	0.96	0.95	0.97	0.94	0.98	0.97	1.00	1.00	0.90	0.94

RC= Recall, F1=F1 Score.

**Figure 6.** Accuracy and precision comparison for all datasets for k-fold method.

competitive performance under k-fold cross-validation, especially on IoTNetIDS, Biflow-MQTT and BiFlowDoS datasets, where accuracy and precision values are close to 1.00. This implies that encoder-based approaches can be applied to both structured and unstructured data. However, for datasets such as DARPA and CAIDA, precision drops to around 0.89 and 0.95, respectively, indicating minor classification challenges under specific conditions.

Figure 7 shows that all the machine learning techniques perform well on different datasets for k-fold method as well. kNN has outstanding AUC at many datasets, such as ToN_IoT and UniflowDoS, by showing its ability to classify one class from another. Deep learning models, such as the ANN and Auto-Encoder, demonstrated better consistency in most data sets, while considering all the ROC curves obtained under the k-fold cross-validation strategy. The Naive Bayes

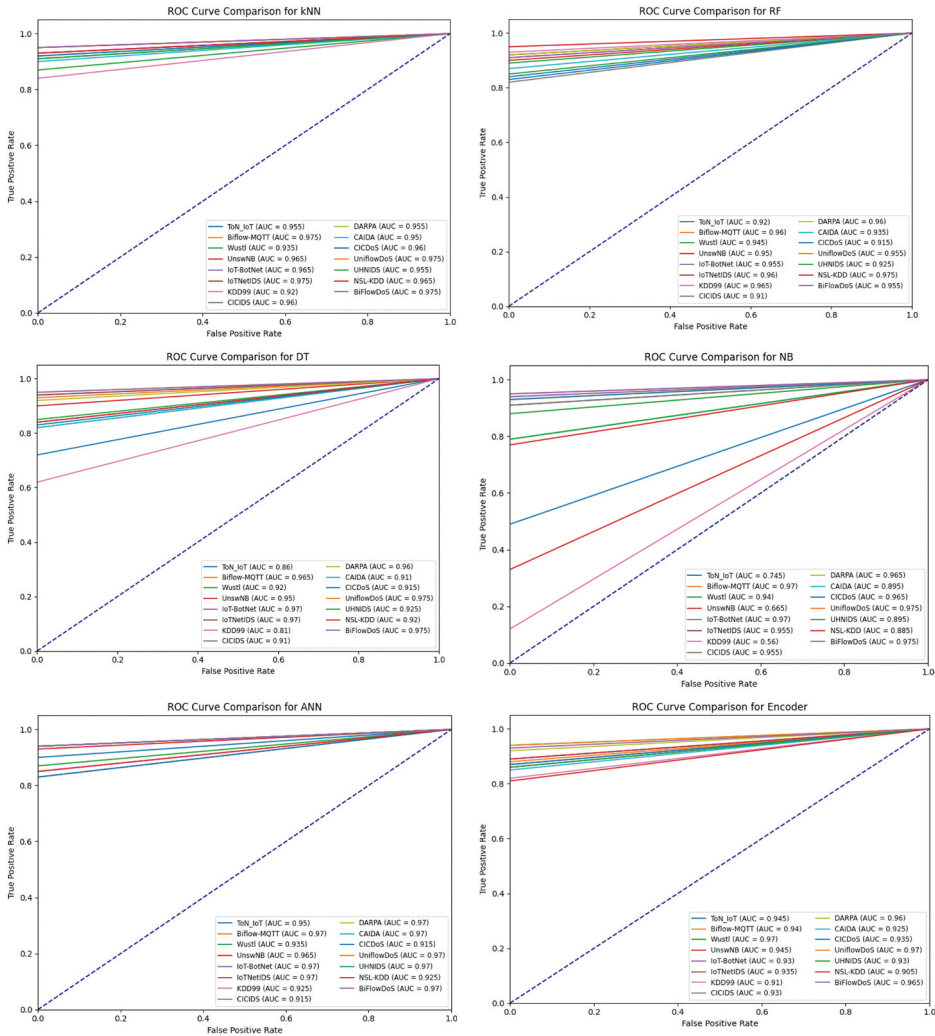


Figure 7. ROC comparison for all datasets and algorithms for k-fold method.

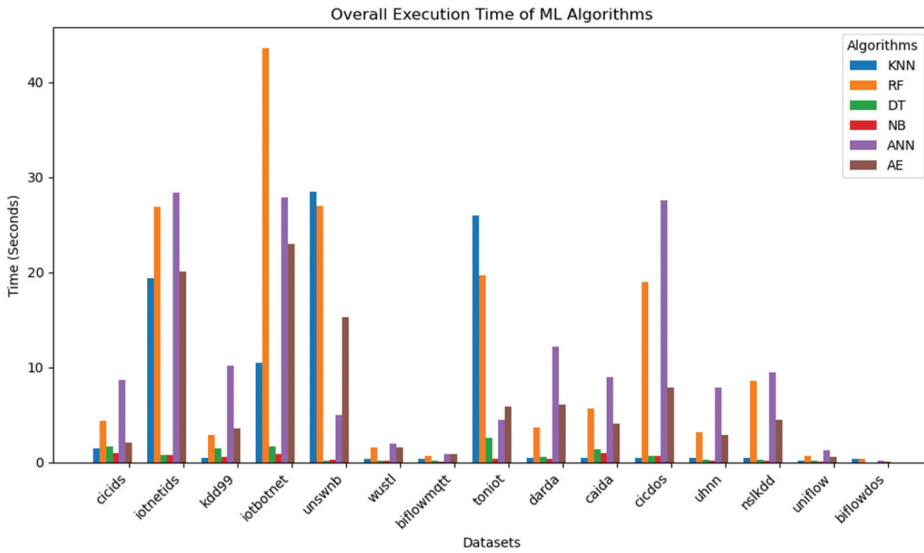


Figure 8. Overall execution comparison for all datasets and algorithms for k-fold approach.

algorithm performs the least, and especially for such databases, like KDD99 or UnswNB, on which the accuracy as well as the precision has sharply decreased.

Figure 8 presents overall execution time for the k-fold validation method. KNN and RF consume more time compared to hold-out approach for larger datasets like iotnetids and ciccids, as these algorithms become computationally intensive in the presence of high dimensions.

kNN and ANN are shown to be the most accurate algorithms in all the given IoT datasets because of their high accuracy, precision, recall and F1 values with strong handling of both the balanced and imbalanced data set. kNN performs well with ToN_IoT, IoTNetIDS and Biflow-MQTT datasets with excellent anomaly detection capabilities and strong AUC scores, but its time complexity, especially during testing time, is a challenge that makes it less scalable in resource-constrained IoT environments. ANN, however, proves to be highly adaptive and, as such, performs outstandingly in both structured datasets such as IoTNetIDS and Biflow-MQTT and complex scenarios such as CICIDS with robust recall and F1 scores to minimize false negatives. ANN is highly resource-intensive at the training stage due to its complex architecture but remains quite efficient at the testing stage, which makes it apt for real-time IoT applications. Overall, kNN and ANN are better candidates for IoT anomaly detection, but considerations of computational complexity and scalability are essential for practical deployment. IoT networks have heterogeneous elements with very dynamically changing attacks vectors, where it is probable that datasets would not account for all of them; therefore, accurate models like kNN and ANN could fall back behind since they are data-dependent for generalization. They must incorporate more adaptive mechanisms, online learning capabilities and adversarial robustness mechanisms.

6. Conclusion

This paper presents a detailed analysis on the performance evaluation of 15 datasets containing traffic data from IoT environments. We used supervised and semi-supervised algorithms such as K-Nearest Neighbour (kNN), Random Forest (RF), Decision Tree (DT), Naïve Bayes (NB), Artificial Neural Network (ANN) and Auto Encoder (AE), which are popular machine learning and deep learning algorithms. We used several performance metrics such as Accuracy, Precision, Recall, F1 Score, ROC and execution time to deeply analyze and discuss the factors influencing the models' performance. Our findings indicate that KNN and ANN exhibited the highest accuracies of 97% and 96%, respectively. Although DT and NB exhibited smaller execution times compared to the others, their performance metrics were poor compared with those of other machine learning algorithms. In fact, while small execution times are a plus factor, kNN and ANN can be regarded as the most effective options to implement for IDS because of their higher accuracy. Future work should consider the evolving threats of IoT, scalability issues and practicality in real-world implementations by focusing on adaptive models that include real-time learning, handle data imbalance and support resource-constrained environments. Exploring computational complexity and proposing novel strategies for scalability and real-time processing in various IoT scenarios could also be important contributions toward furthering IDS effectiveness in addressing the modern IoT security challenges.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Availability of data and material

The datasets used in this study are publicly available.

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